Making Money While You Sleep?

Anomalies in International Day and Night Returns

Kevin Aretz* **and Söhnke M. Bartram†**

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In this paper, we decompose return premia into day and night components based on a sample of more than 48,000 stocks from 35 countries including the United States. Day returns are higher than night returns, but have similar volatility. Payoffs to value- or equally-weighted investment strategies based on size, book-to-market, momentum, long-term reversals, accruals, net share issuances, asset growth, and profitability tend to be positive for day returns, but negative or zero for night returns. Fama-MacBeth tests show that the well-known relationships between total returns and lagged firm characteristics are the result of their relationships with day returns, while their relationships with night returns are of opposite signs. While contemporaneous, independent work by Lou et al. (2015) also documents that U.S. price momentum occurs overnight, this result is the exception rather than the rule, since price momentum as well as most other anomalies are driven by day returns outside the United States. Results are robust across countries and subsamples, although there are variations in the magnitude of the strategy payoffs across countries.

Keywords: Overnight returns, asset pricing, return factors
JEL Classification: G3, F4, F3
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1 Introduction

How do market closures affect the mean and standard deviation of returns? Market closures preclude investors from trading in the market as well as from learning about the economy by observing market prices and trading activities, changing the timing of the resolution of uncertainty and shifting risk across periods and across agents. While theoretical models predict either higher or lower returns and volatilities during trading periods (e.g., Foster and Viswanathan, 1990; Slezak, 1994; Longstaff, 1995; Hong and Wang, 2000), empirical studies typically find higher return volatility when markets are open (e.g., Christie, 1981; French and Roll, 1986), but no clear evidence of day returns being either higher or lower than night returns (e.g., Rogalski, 1984; Wood et al., 1985; Harris, 1986; Smirlock and Starks, 1986).

In this paper, we contribute to the literature by conducting a comprehensive analysis of the effect of market closures on stock returns. Our paper is first to analyze this phenomenon internationally, based on a sample of more than 48,000 stocks from 35 countries around the world (including the United States) that spans the period from 1993 to 2011. Moreover, it extends the literature by decomposing not only returns but also the payoffs associated with eight common financial market anomalies into day and night components. In particular, we calculate returns and return volatilities for full trading days (total returns) as well as for trading periods (day returns) and non-trading periods (night returns). Our results show that day returns tend to exceed night returns, but that the two have a similar level of volatility. To wit, day returns are on average larger (10.50% p.a. vs. 5.43% p.a.), but have a similar standard deviation (8.21% p.a. vs. 8.16% p.a.) compared to night returns. However, there are significant variations across countries, with about the same number of countries exhibiting a significantly positive and negative difference between day returns and night returns, respectively. Thus, the international evidence is consistent with the mixed nature of the existing empirical evidence for the United States.

More importantly, we also calculate the returns of quintile spread portfolios formed according to investment strategies based on size, book-to-market, momentum, long-term reversals, accruals, net share issuances, asset growth, and profitability and decompose their returns into day and night components. We show that the strategies tend to produce positive day returns, but negative or zero night returns. The conclusions hold for equally-weighted and value-weighted portfolios. Fama-MacBeth regressions show that the well-known relationships between total returns and lagged firm

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characteristics are the result of their relationships with day returns, while their relationships with night returns are of opposite signs. Similar to contemporaneous, independent work by Lou et al. (2015), we document that U.S. price momentum occurs overnight, but this result is the exception rather than the rule, since price momentum as well as other anomalies are often driven by day returns outside the United States. Our results are robust across countries and various subsamples, but they vary in magnitude across countries.

The remainder of this paper is structured as follows. The next section provides an overview of theoretical predictions regarding day and night returns and related literature. Section 3 describes the sample and data sources. Section 4 outlines the different tests and reports our results. Section 5 concludes.

2 Theoretical Predictions and Empirical Evidence

Many theoretical predictions in the day and night return-literature are based on the incorporation of information in stock prices. During trading hours, public information is more likely to arrive given the overlap with business hours and lower information collection costs. Also, during trading hours, information accumulated during market closure periods and trading by informed investors is incorporated into prices and might lead to pricing errors. All these observations suggest higher volatility during trading hours (e.g., see French and Roll, 1986; Admati and Pfleiderer, 1989; Foster and Viswanathan, 1990; Hong and Wang, 2000). To illustrate, a market closure delays the release of private information, subjecting uninformed agents to greater risk from adverse selection in the post-closure market (Slezak, 1994). Moreover, closures also affect the variance of returns by altering both the pre- and post-closure liquidity costs, and depending upon the degree to which prices reflect pre-closure news, the risk associated with providing liquidity may increase or decrease, which can lead to higher or lower volatility during trading hours.

The fact that investors need to be compensated for the risk of holding positions during market closures without being able to liquidate them suggests that night returns should be higher than day returns (e.g., Longstaff, 1995). Mean returns, however, depend upon the change in the level of risk, which, depending upon the percentage of informed agents in the market, can be either more or less than without market closure (Slezak, 1994). When there are few informed agents, prices reflect very little current information, and closures shift most of the risk associated with closure news to the post-closure market, significantly increasing the uncertainty in the post-closure market and reducing

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the amount of uncertainty resolved over the closure. Consequently, mean returns over closures are smaller than they would be if the market had not closed.

Since the lack of overnight trading opportunities increases the risk of overnight stock positions, investors reduce their hedging trades (trades that hedge the risk of illiquid assets) at the market close (Hong and Wang, 2000). The anticipation of this decrease in hedging trades leads to a decrease in stock prices as the close approaches since the risk premium increases. At the same time, since market prices are not available as a source of information during market closures, information asymmetry among investors increases with market closures. Information asymmetry, however, tends to decrease as markets reopen, reducing the premium and increasing the price of stocks. The relative strength of both effects determines the level and volatility of returns. For example, when decreasing information asymmetry leads to increasing stock prices during the day, returns are higher over trading periods, while they are higher over non-trading periods in case of symmetric information.

Empirical studies, typically carried out on U.S. data, suggest that day returns are more volatile than night returns (e.g., Fama, 1965; Granger and Morgenstern, 1970; Oldfield and Rogalski, 1980; Christie, 1981; French and Roll, 1986). In contrast, the evidence with regards to mean returns is conflicting, with some studies reporting higher (Wood et al., 1985) and others reporting lower overnight returns (Rogalski, 1984; Harris, 1986). Smirlock and Starks (1986) show that night returns are lower (higher) during the 1974-1983 (1963-1968, 1968-1974) period.[3] Recent evidence shows that mean returns generated during market opening hours contribute much less to mean total returns compared to mean overnight returns (e.g., see Cliff et al., 2008). A possible explanation for this result is that retail investors, who are expected to dominate trading at the open, inflate the opening prices of stocks that have recently attracted their attention (see Berkman et al., 2012).

3 Sample and Data

3.1 Total, Day, and Night Returns

We calculate the net return of stock i over trading day t's day period, $r_{i,t}^{Day}$, as:

$$r_{i,t}^{Day} = (P_{i,t}^{Close} + Div_{i,t} - P_{i,t}^{Open})/P_{i,t}^{Open}, \quad (1)$$

[3] French (1980), Gibbons and Hess (1981), and Keim and Stambaugh (1984) document negative returns on Monday.

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where $P_{i,t}^{Open}$ and $P_{i,t}^{Close}$ are the opening price and the closing price, respectively, and $Div_{i,t}$ is the total dividend paid by stock i. Both the opening price and the closing price are trade prices, and not quotes. When $P_{i,t}^{Open}$ is missing because there is no trading at the open, we replace it with the stock's closing price at the end of the previous trading day. We winsorize day returns and total (i.e. daily) returns at the 0.1^{th} and 99.9^{th} percentiles. Using the day return, we back out the night return, $r_{i,t}^{Night}$, as:

$$r_{i,t}^{Night} = \left(\frac{1+r_{i,t}^{Total}}{1+r_{i,t}^{Day}} \right) - 1. \tag{2}$$

3.2 Firm Characteristics for Investment Strategies

We decompose the payoffs accruing to investment strategies based on size, book-to-market, momentum, long-term reversals, accruals, net share issuances, asset growth, and profitability into their day and night components. In doing so, we follow Fama and French (2008) in calculating the firm characteristics. In particular, at the end of June of each year *t*, we take the contemporaneous values of all market variables required to construct the firm characteristics. On the same date, we also take the values of all accounting variables required to construct the firm characteristics from the fiscal year that ended during the last calendar year (or earlier). We keep these values from July of year *t* to June of year *t+1* to ensure that the firm characteristics could have been observed by investors in calendar-time. The two exceptions are the momentum past return and the long-term reversal past return, which are updated on a monthly basis.

We define size as the log of the product of share price times common shares outstanding. Book-to-market is the book value of equity divided by market capitalization. The book value of equity is defined as total assets minus the sum of total liabilities, deferred taxes, and the value of preferred stocks. The value of preferred stocks is either their liquidating, redemption, or capital value (in this order). We set deferred taxes and the value of preferred stocks equal to zero if they are missing. The momentum past return is the compound return over the previous twelve months, excluding the most recent past month. We exclude the most recent month to eliminate short-term reversal effects. The long-term reversal past return is the compound return over the previous 36 months, excluding the most recent twelve months.

Net share issuances is the log of the ratio of the most recently observed number of shares-value to its one year-lagged value, where the number of shares are adjusted by multiplying them by

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the number of shares-adjustment factor. We set net share issuances to zero if the date associated with the lagged value occurs before the date on which the stock first appears in CRSP (U.S. stocks) or Datastream (non-U.S. stocks). Accruals is the spread between the most recently observed working capital value and its one year-lagged value, scaled by the book value of equity. We define working capital as current assets minus the sum of cash and cash equivalents and current liabilities minus debt in current liabilities, and we scale working capital by the adjusted number of shares. Asset growth is the log of the most recently observed total assets value and its one year-lagged value, where both the current and lagged value are deflated by the number of shares-adjustment factor. Profitability is income before extraordinary items minus preferred dividends and deferred income taxes, scaled by the book value of equity. We set preferred dividends and deferred income taxes to zero if they are missing. We winsorize the firm characteristics at the 0.5^{th} and 99.5^{th} percentiles.

We sort stocks into portfolios according to the quintile breakpoints of the firm characteristics. Even when we use global data to create the portfolios, we calculate the quintile breakpoints separately by country and month, so that each country contributes a representative number of stocks to each global portfolio. For simplicity, we put those stocks expected to produce the higher mean returns into the higher quintile-value portfolios. To give an example, size quintile 1 contains the largest stocks in each country, whereas size quintile 5 contains the smallest stocks. We either value- or equally-weight the characteristic portfolios.

3.3 Data

Our sample consists of 48,413 stocks from 35 countries. Our data period ranges from January 1993 to January 2012. We start the data period in 1993 because opening prices are only available from this date on in our data sources. We source U.S. stock market data (i.e., stock prices, returns, dividends, and adjustment factors) from CRSP and U.S. accounting data from Compustat. For non-U.S. firms, we source market data (in U.S. dollars) from DataStream and accounting data (in local currencies) from WorldScope. For robustness tests, we also source U.S. pricing data from TAQ. We define the TAQ opening price as the price of the first trade with a positive size occurring after 9 am. We include all publically traded common stocks, except those associated with firms operating in the financial industry (SIC Code: 6000-6999) and those with a negative book value of equity.

We only analyze U.S. stocks that are traded on the NYSE, Amex or Nasdaq, and we exclude U.S. stocks with a one month-lagged price below $2. In case of the other countries, we include stocks from all exchanges. Because non-U.S. data are often of a lower quality than U.S. data, we im-

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pose some additional data filters on these data. In particular, we set to missing observations associated with extreme price reversals over either one trading day (intraday) or two consecutive trading days. We define an intraday price reversal as a day or night return above 100%, but a total return below 10%. We define a two-day reversal as a total return over any of the two days above 100%, but a compound return over the two days below 10%. We drop stocks that are not traded during more than 30% of all trading days over the previous one year. We also apply additional filters to eliminate obvious errors as, for example, when the opening price is denominated in another currency than the closing price. Finally, we exclude countries with less than 30 firms on average or less than eight years of market data or accounting data.[4]

4 Empirical Results

4.1 Day and Night Returns

Table 1 provides descriptive statistics on raw returns, calculated either over the entire day (from close to close, i.e., total daily returns) or over the day (open to close) or night (close to open) periods. Panel A shows statistics aggregated across countries, where averages are calculated by first averaging by country and subsequently across countries, whereas Panel B reports statistics by country. Across countries, the annualized total return over the sample period is 12.3%, with a volatility of 13.8%. Decomposing total returns shows that most of the total return is attributable to the day period. In particular, the annualized day return is 10.5% p.a., while the annualized night return is 5.4% p.a. The difference of around 5% is highly statistically significant (p-value = 0.01). In contrast, the volatilities of day returns and night returns are similar. To wit, the volatility of the global day return is 8.21%, whereas the volatility of the global night return is 8.16%, which are indistinguishable at conventional significance levels.

Breaking the results down by day of the week shows that there are strong discrepancies in returns at the beginning and at the end of the week. In particular, while overnight returns vary less over the week, day returns are negative at the start of the week, but positive at its end. Thus, the difference between day and night returns is negative and highly significant at the start of the week, but positive and highly significant at its end. The above patterns are surprising, given that day periods are of equal length and might be expected to have similar amounts of information dissemination and

[4] We also exclude Greece, Pakistan, and Sweden because these countries tend to produce negligible day returns, stemming from the fact that their opening prices are often identical to their closing prices.

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trading. In any event, it appears that investors are compensated for holding stocks in both day and night periods. Day returns are larger for most months throughout the year and in the first half of our sample period, though the difference is not always significant.

In line with prior evidence, U.S. day returns are more volatile than U.S. night returns (e.g., Fama, 1965; Granger and Morgenstern, 1970; Oldfield and Rogalski, 1980; Christie, 1981; French and Roll, 1986). However, day returns are also larger than night returns in the United States, which can be in contrast to prior evidence (Rogalski, 1984; Harris, 1986; Smirlock and Starks, 1986). Internationally, there are an almost equal number of countries in which day returns exceed night returns and vice versa, and the variations across countries are not entirely attributable to variations in developmental status. To wit, the mean spread between day and night returns is positive and statistically significant (at the 10% significance level or better) for *both* stocks in developed countries and stocks in developing countries.

Figure 1 plots the cumulative total return, day returns and night return over our sample period. While the cumulative returns do not correspond to realistic investment strategies (because they would require a daily turnover of 200%), they are informative about how prices move over the different trading periods. The figure suggests that day returns outperform night returns, as shown in Panels A, B, and C for the World, the United States, and other (non-U.S.) countries, respectively.

4.2 Portfolio Sorts

Our next aim is to assess to what extent the risk premia associated with the firm characteristics originate from day or night returns. We do so as follows. At the end of every month, we sort the stocks in each country into quintile portfolios according to the lagged firm characteristic values and calculate the average return of the stocks in the next period.[5] We also calculate the spread in returns between the extreme portfolios and test whether it differs from zero. Panel A of Table 2 shows results for equally-weighted portfolios sorted according to size, book-to-market, momentum, long-term reversal, accruals, net share issuances, asset growth, and profitability. For each characteristic, we study the total return performance of the portfolios to validate that the firm characteristics yield return premia in line with the literature.

[5] The firm characteristic values are calculated in such a way that they could have been observed by investors at the time (see Section 3.2).

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We find this to be the case, as the payoffs to all eight investment strategies are positive and statistically significant. Momentum and value strategies have particularly large premia. Analyzing the day and night performance of the investment strategies separately, we find that the premia of the investment strategies are large and positive during the day, but typically negative overnight. The only exception is accruals, for which day and night returns are positive and of about the same (small) magnitude. Panel B shows results for value-weighted portfolios. While there are some variations in magnitudes, the overall thrust of the results is the same. Thus, most stock market anomalies appear to occur during the day trading period and reverse over the night period.

In Figure 2, we plot the performance of the investment strategies by country. Each panel shows the average spread portfolio return for one of the eight anomalies, with separate bars for day and night returns. There are two take-aways. First, the spread portfolios tend to produce negative returns at night (bars in light color), while they tend to produce positive returns during the day (bars in darker color). Hence, the aggregate results shown in Table 2 are not driven by single countries, but are reflective of a general pattern across countries. Second, there are significant variations in the magnitude of the payoffs, both across strategies and, more importantly, across countries. While payoffs to the momentum strategy are known to be large, the figure also shows large size and profitability premia in some countries. In contrast, payoffs to accruals, asset growth, or net share issuance strategies are typically an order of magnitude smaller.

As in contemporaneous, independent work by Lou et al. (2015), payoffs to a momentum strategy based on U.S. stocks are large and positive at night, while they are small and negative during the day (Panel C). Nevertheless, our results suggests that the United States is atypical in this respect, as the payoffs to momentum strategies in countries outside the United States tend to occur during the day. Similarly, the payoffs to our other style strategies based on U.S. stocks, such as the size, value, reversal, accruals, net share issuance, asset growth, and profitability strategy, occur primarily during the day, and this appears to be the picture across countries more generally.

Figure 3 shows the cumulative investment performance for equally-weighted and value-weighted portfolios of day and night return based on the eight anomalies. Consistent with Figure 2, it demonstrates the outperformance of day return strategies over those based on night returns. The figure also points to variations in the economic magnitude of the effects. For example, the size and the momentum strategies produce economically significant effects, while payoffs to strategies based on accruals, asset growth, and profitability are much smaller. While these analyses are informative,

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they are limited in that they are univariate (i.e., they do not control for other effects). In order to assess how different drivers of returns relate to returns in the presence of other factors, a multivariate analysis is required, as conducted in the next section.

4.3 Fama-MacBeth Regressions

We next assess to what extent the firm characteristics predict stock returns in a multivariate setting. In doing so, we conduct Fama-MacBeth regressions of next month's return (compounded to the monthly level from daily total returns, day returns or night returns) on a comprehensive set of firm characteristics. Regressions with monthly total returns correspond to analyses that have been commonly employed in the prior literature. The first three columns in Panel A of Table 3 show results for firms from all countries (World). The results confirm many well-documented relationships between future returns and firm characteristics. Book-to-market, momentum, and profitability have positive coefficients; size, accruals, share issuances, and asset growth have negative coefficients. Several effects are significant. Our conclusions continue to hold when we alternatively use samples of U.S. or non-U.S. firms, with some variations in economic magnitude and significance levels.

Panel A also shows results for regressions with day and night returns as dependent variable. These regressions investigate to what extent the relationships for total returns are driven by their day and night components. Results show that stock market anomalies are the result of how the underlying characteristics relate to day returns, while the relationships with night returns are often of the opposite sign. In particular, size, share issuance, and asset growth are negatively related to day returns, while book-to-market, momentum, and profitability have positive relationships with returns. In contrast, the coefficients of these variables are of opposite signs for night returns. Results are robust to splitting the sample into U.S. and non-U.S. subsamples of firms.

Panel B shows the results from Fama-MacBeth regressions run separately for the eight most important non-U.S. countries (Australia, Canada, France, Germany, Hong Kong, Italy, Japan, and the U.K.). While we find some variations in strategy payoffs across countries, the general pattern is that most anomalies arise during the day period, but reverse at night. Exceptions are size in Australia, momentum in Hong Kong, and asset growth in Japan.

We conduct several additional tests to confirm the robustness of our results. In particular, we verify the opening prices provided by CRSP by benchmarking them against opening prices provided by TAQ. For each U.S. stock and date, we download the TAQ price of the first trade with a

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positive volume occurring after 9:45 am. We subsequently limit the sample to observations where the difference between the TAQ price and the CRSP opening price is less than 10% of the CRSP closing price. The results of the portfolio exercises and the Fama-MacBeth (1973) tests for these observations are virtually identical to those reported in Tables 2 and 3.

5 Conclusion

This paper is the first to analyze day and night returns internationally. Our results show that day returns exceed night returns on aggregate, but have similar volatility. We also find evidence that the returns on strategies based on size, book-to-market, momentum, long-term reversal, accruals, share issuance, asset growth, and profitability are driven by day returns, while their returns are often negative or zero during night periods, both for equally-weighted and value-weighted portfolios. Fama-MacBeth tests show that the well-known relationships between future total returns and the firm characteristics are the result of their relationships with day returns, while relationships with night returns are of opposite signs. Results are robust across countries and various subsamples.

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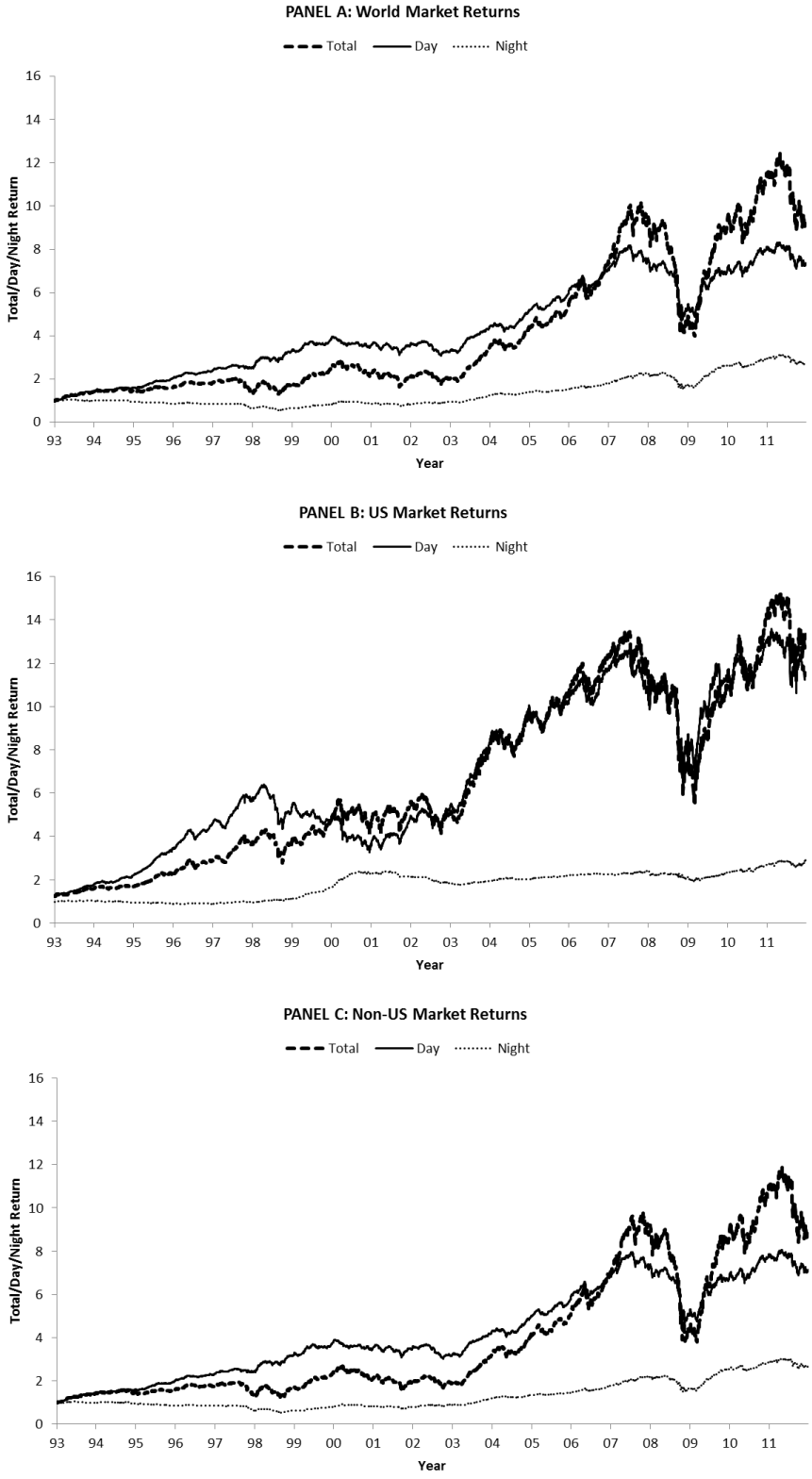
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Figure 1: Market Returns

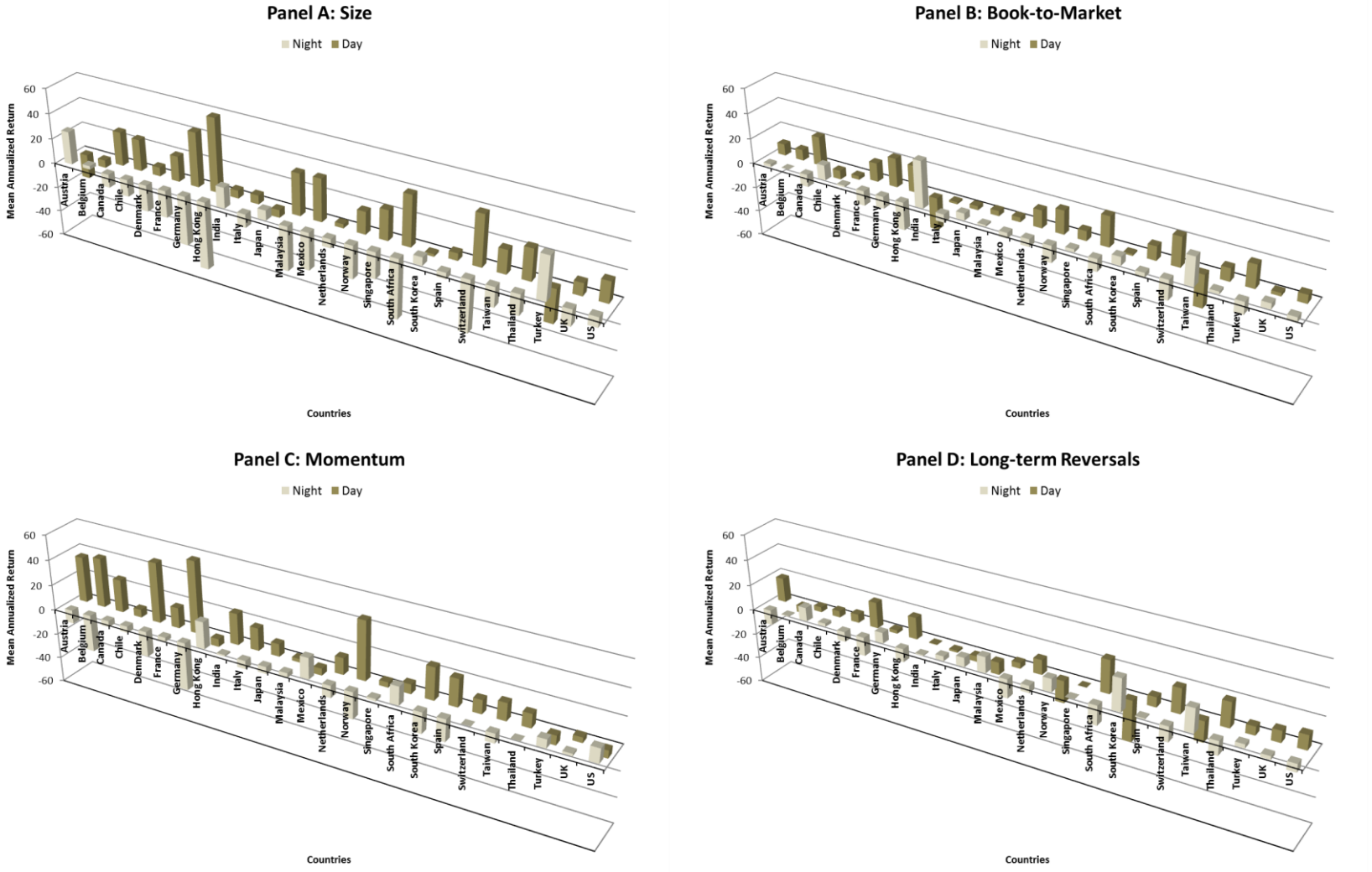
The figure shows the cumulative total returns, day returns and night return for all markets (Panel A), the U.S. market (Panel B), and non-U.S. markets (Panel C). To calculate cumulative returns in Panels A and C, we first compute equal-weighted averages by country and subsequently average across countries.



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Figure 2: Anomaly Investment Strategies By Country

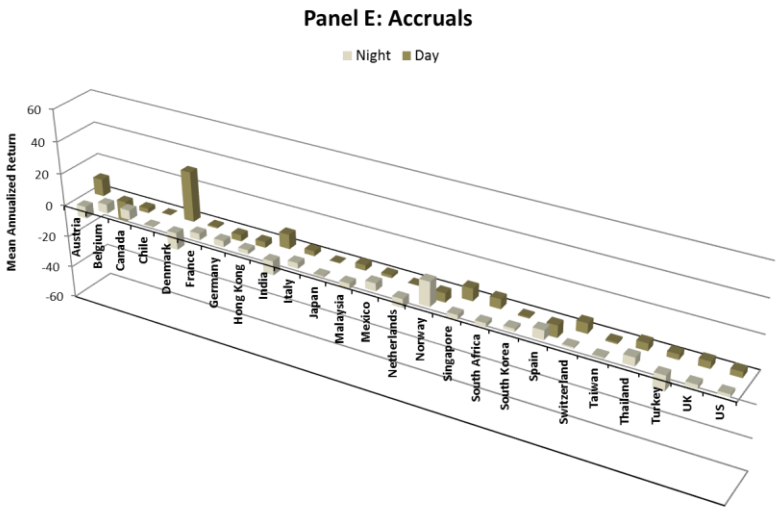
The figure shows the mean return of country-specific quintile spread portfolios formed according to investment strategies based on size (Panel A), book-to-market (Panel B), momentum (Panel C), long-term reversal (Panel D), accruals (Panel E), net share issuance (Panel F), asset growth (Panel G), and profitability (Panel H). The mean return is annualized (by multiplying the daily return by 252) and in percentage. The bars in the front row (in light grey) show the spread portfolio return over the night period, the bars in the back row (dark green) the spread portfolio return over the day period. The spread portfolios are set up in such a way that they are expected to yield a positive mean return.

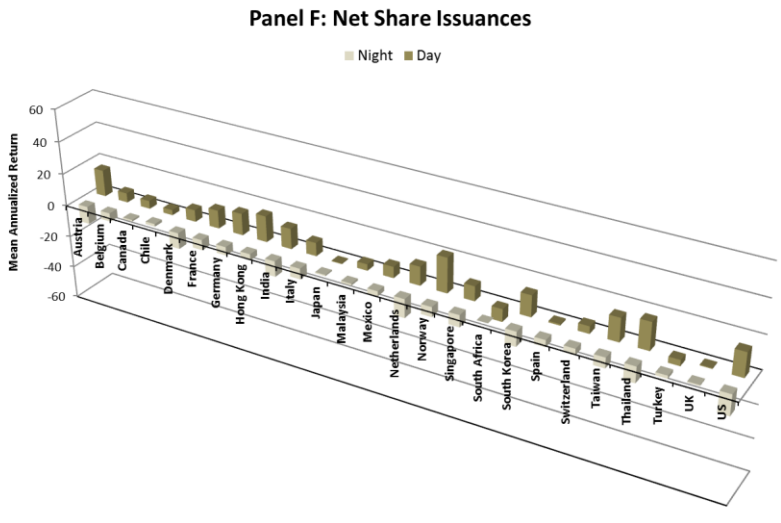


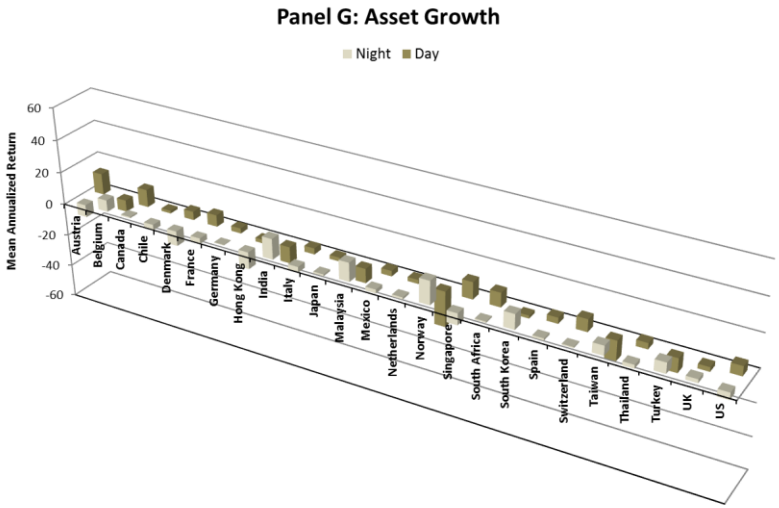
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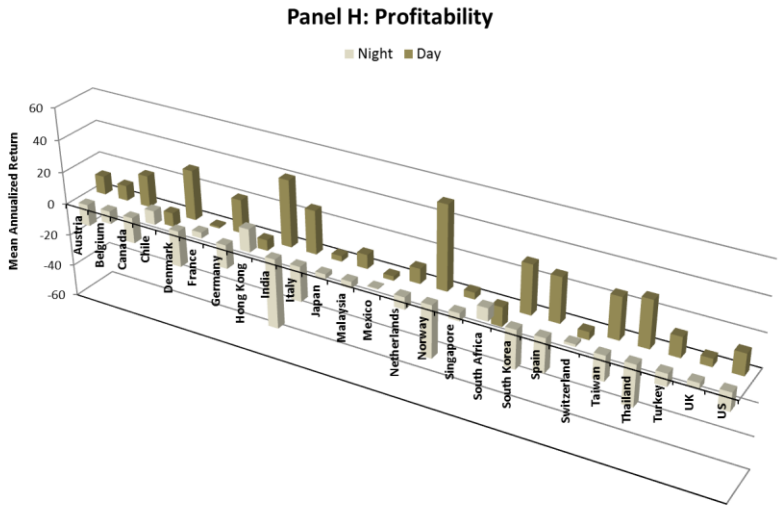
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Figure 2: Anomaly Investment Strategies By Country (continued)







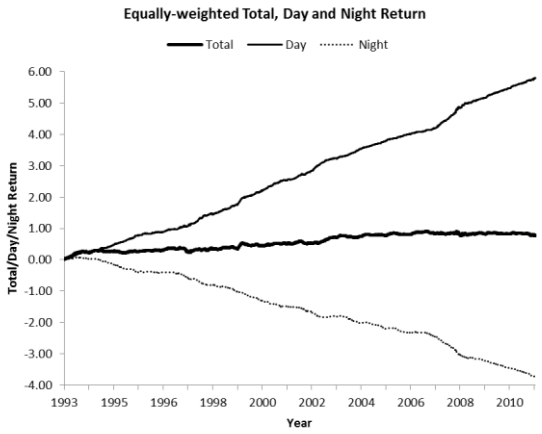


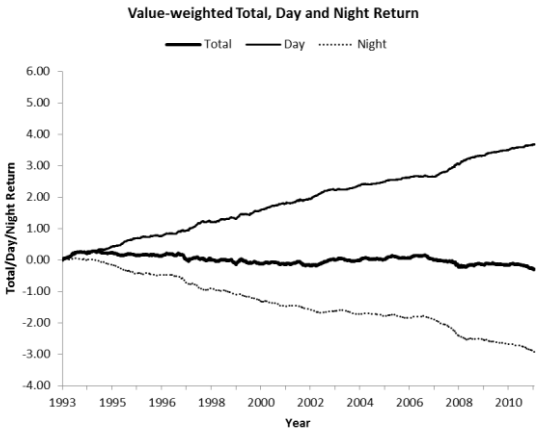
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Figure 3: Cumulative Anomaly Performance

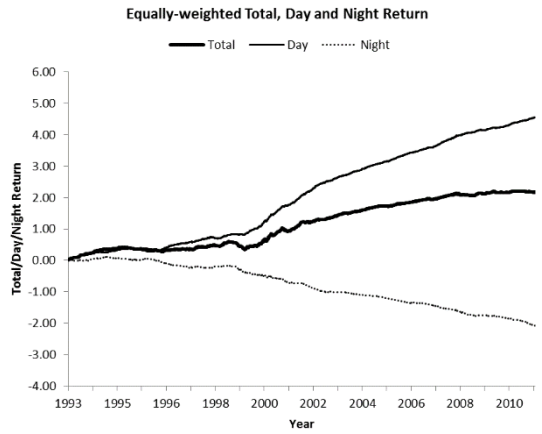
The figure shows the cumulative return of equally- or value-weighted spread portfolios formed based on size (Panel A), book-to-market (Panel B), momentum (Panel C), long-term reversal (Panel D), accruals (Panel E), net share issuance (Panel F), asset growth (Panel G) and profitability (Panel H). The fat bold line is the total return, the thin bold line the day return, and the dashed thin line the night return. To calculate the compound return, we first take equal-weighted (left panels) or value-weighted (right panels) averages of returns by country and subsequently average across countries. The spread portfolios are constructed in such a way that they are expected to yield a positive mean return. The definitions of the anomaly variables are provided in Section 3.1.

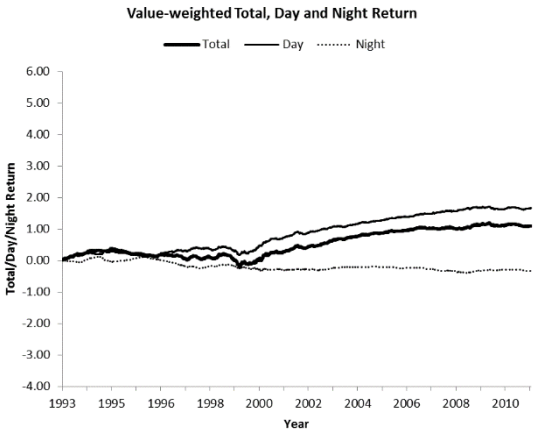
Panel A: Size





Panel B: Book/Market



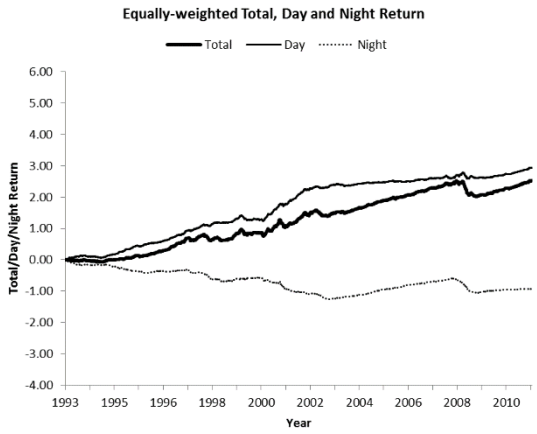


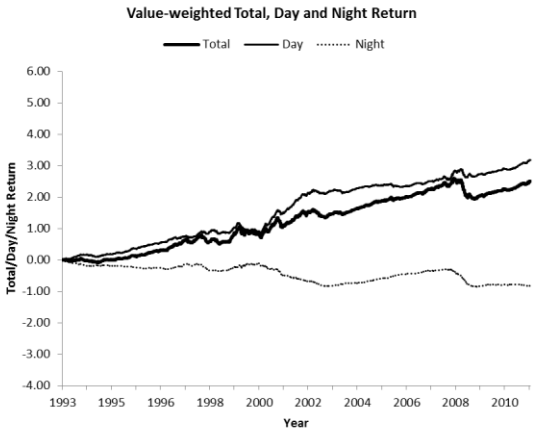
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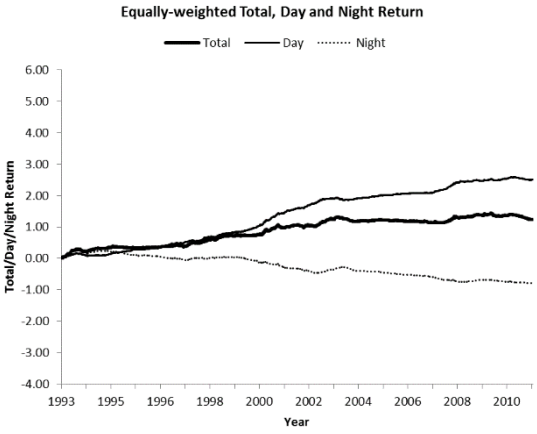
Figure 3: Cumulative Anomaly Performance (continued)

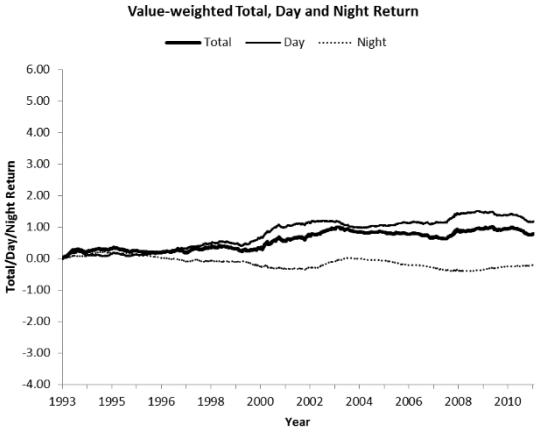
Panel C: Momentum



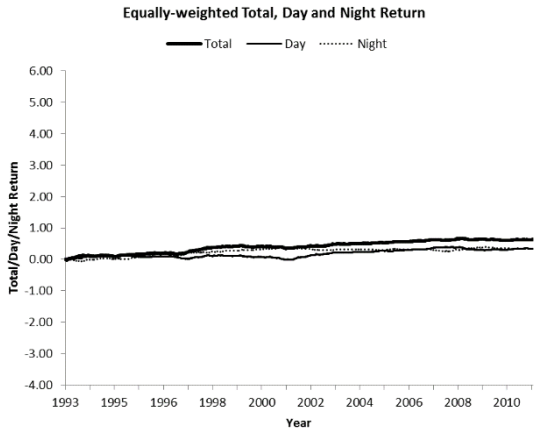


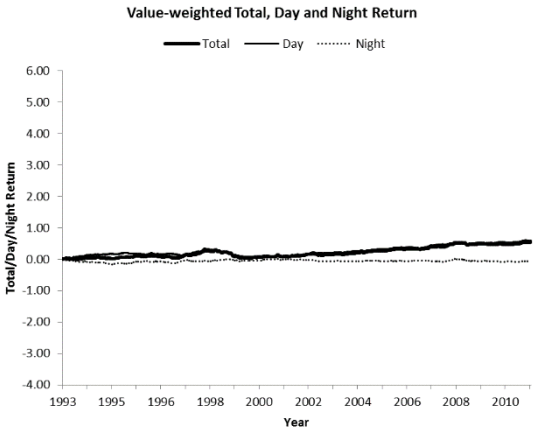
Panel D: Long-term Reversal





Panel E: Accruals



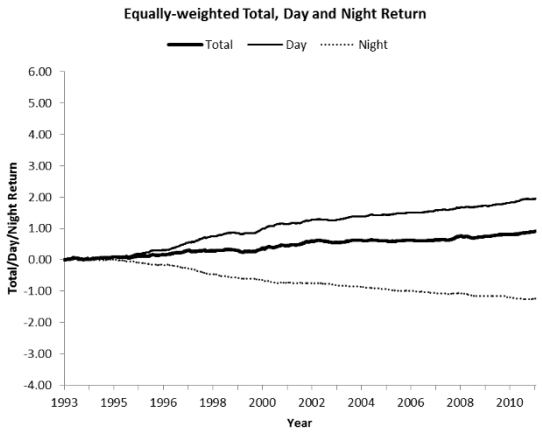


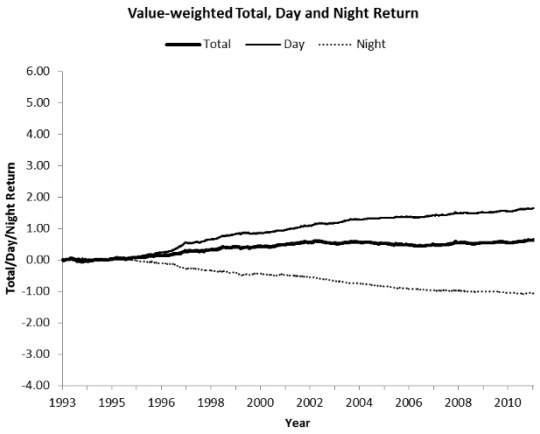
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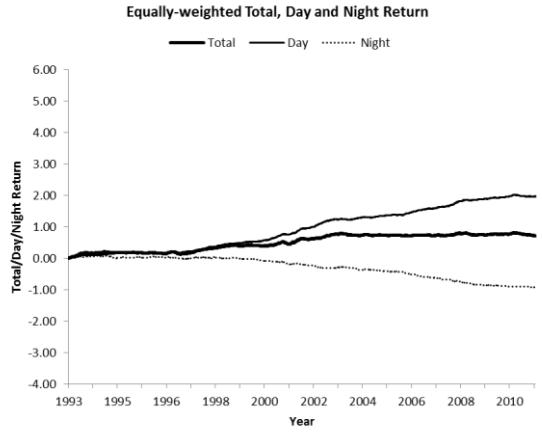
Figure 3: Cumulative Anomaly Performance (continued)

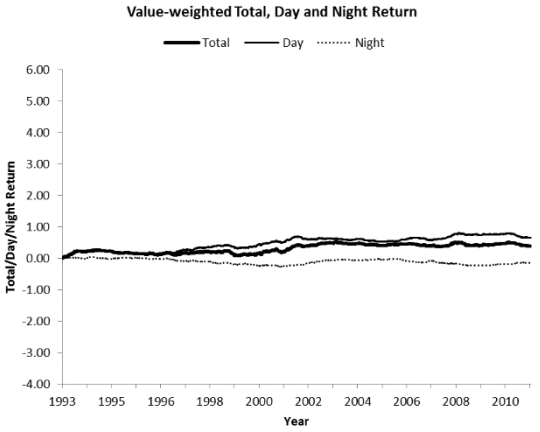
Panel F: Net Share Issues



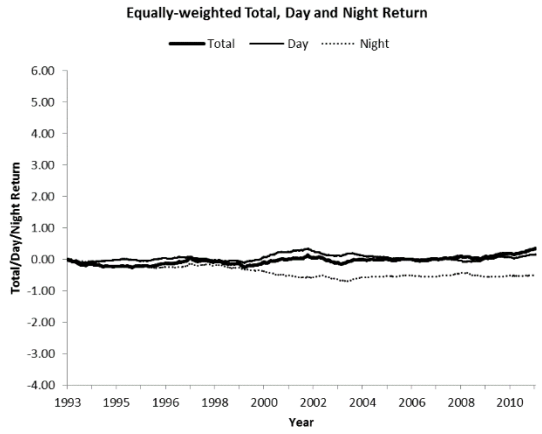


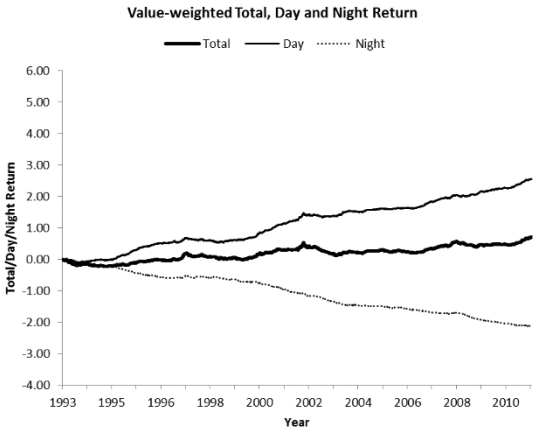
Panel G: Asset Growth





Panel H: Profitability





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Table 1: Annualized Total Returns, Day Returns and Night Returns

The table shows the mean and standard deviation (St.Dev.) of annualized total returns, day returns, and night returns (in percentage). We annualize returns by multiplying the mean by 252. The last columns report statistics on the difference between day returns and night returns as well as the *p*-value from *t*-tests analyzing whether day returns are statistically different from night returns. To compute the table entries, we first average returns by country and subsequently across countries. Panel A shows summary statistics for all observations and by weekday, calendar month, and year. Panel B shows summary statistics by country and developed and developing countries.

Panel A: Aggregate Statistics

	Total Return		Day Return		Night Return		Day - Night		
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	t-Test	Std. Dev.
Pooled	12.30	13.83	10.50	8.21	5.43	8.16	5.06	0.01	0.05
Day of the week									
Monday	-8.33	16.58	-9.50	9.47	4.59	8.82	-14.09	0.00	0.65
Tuesday	-5.28	12.56	-0.58	7.82	-0.81	8.10	0.23	0.96	-0.28
Wednesday	22.67	13.47	17.03	8.27	9.29	8.19	7.74	0.11	0.08
Thursday	15.14	13.84	9.53	8.25	9.01	7.79	0.51	0.90	0.46
Friday	37.28	12.09	36.01	6.70	5.08	7.84	30.92	0.00	-1.13
Month									
January	21.52	13.79	14.17	8.44	10.69	7.87	3.48	0.61	0.57
February	22.91	11.59	11.23	7.11	14.94	6.89	-3.71	0.56	0.23
March	24.85	13.64	15.10	8.38	13.23	7.54	1.87	0.77	0.84
April	47.20	11.91	30.64	7.21	19.86	7.18	10.78	0.09	0.03
May	9.99	13.73	9.70	8.58	3.54	7.24	6.16	0.32	1.34
June	-4.60	12.57	-2.58	7.32	1.43	6.97	-4.01	0.45	0.34
July	13.53	11.06	10.19	6.80	7.09	6.33	3.09	0.57	0.47
August	-9.96	13.49	1.66	8.36	-7.85	7.97	9.51	0.18	0.39
September	-13.13	15.64	-0.17	9.12	-8.95	9.74	8.78	0.29	-0.62
October	3.28	19.91	5.03	10.32	2.26	13.00	2.77	0.77	-2.68
November	4.42	14.95	4.06	8.97	4.23	8.40	-0.17	0.98	0.56
December	28.12	10.82	26.71	6.84	5.50	6.24	21.21	0.00	0.61
Year									
1993	34.08	8.34	35.68	5.36	0.17	4.25	35.51	0.00	1.11
1994	3.97	8.59	10.50	5.83	-3.74	4.97	14.24	0.03	0.86
1995	9.64	7.32	23.23	4.69	-10.57	4.07	33.80	0.00	0.62
1996	11.75	6.12	15.23	4.28	-0.56	3.29	15.79	0.00	0.99
1997	-21.56	12.23	6.78	6.40	-24.45	9.31	31.23	0.00	-2.90
1998	19.02	17.82	25.23	10.00	-1.37	10.27	26.60	0.01	-0.27
1999	35.95	9.32	15.36	6.61	24.73	5.02	-9.37	0.18	1.59
2000	-11.96	12.15	-8.35	7.26	1.76	8.64	-10.11	0.32	-1.38
2001	-4.83	14.32	1.87	8.87	-1.49	9.06	3.37	0.75	-0.19
2002	-2.33	11.49	-8.52	7.28	11.50	6.92	-20.02	0.02	0.35
2003	51.54	8.46	28.09	5.81	27.47	4.65	0.62	0.92	1.16
2004	25.25	10.14	16.85	6.46	11.77	4.74	5.08	0.31	1.72
2005	18.74	8.17	13.54	4.96	7.65	4.17	5.90	0.15	0.79
2006	32.23	11.90	16.65	7.39	18.03	6.14	-1.38	0.83	1.25
2007	25.50	13.88	6.84	8.68	21.06	8.64	-14.22	0.16	0.03
2008	-65.17	27.13	-33.43	14.25	-25.88	17.71	-7.55	0.65	-3.46
2009	66.14	20.04	27.60	12.48	42.09	9.96	-14.49	0.16	2.52
2010	24.36	16.13	14.92	9.29	11.92	8.30	3.00	0.67	0.99
2011	-18.51	18.38	-8.52	10.81	-6.85	9.92	-1.67	0.86	0.89

(continued)

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Table 1: Annualized Total Returns, Day Returns and Night Returns (continued)

Panel B: Statistics by Country

	Total Return		Day Return		Night Return		Day - Night		
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	t-Test	Std. Dev.
Country									
Australia	12.62	20.78	-9.23	10.96	23.45	12.63	-32.68	0.00	-1.67
Austria	-13.52	27.23	-10.15	16.93	-1.60	18.30	-8.54	0.50	-1.37
Belgium	12.45	18.03	-4.70	11.83	18.15	9.19	-22.85	0.00	2.64
Brazil	42.85	27.48	7.64	16.74	35.93	14.56	-28.30	0.00	2.18
Canada	29.88	19.91	39.65	13.34	-0.85	10.17	40.50	0.00	3.17
Chile	-22.82	28.76	-31.39	21.20	8.30	10.32	-39.69	0.02	10.88
China	16.72	27.92	28.79	26.01	-10.78	10.41	39.57	0.00	15.60
Denmark	2.34	20.78	-7.12	13.78	12.62	11.07	-19.73	0.00	2.71
Egypt	-6.66	30.79	-10.51	17.95	7.25	21.50	-17.76	0.15	-3.55
France	10.25	15.66	13.63	10.00	-1.33	8.70	14.97	0.00	1.30
Germany	3.97	19.31	19.87	14.60	-9.34	9.60	29.21	0.00	5.00
Hong Kong	3.02	25.50	29.54	18.77	-20.26	14.94	49.80	0.00	3.83
India	22.65	26.39	-28.29	18.50	71.48	16.23	-99.77	0.00	2.27
Indonesia	20.18	36.64	36.88	26.47	-14.24	21.63	51.11	0.00	4.84
Israel	17.66	22.75	6.12	16.51	14.72	10.36	-8.61	0.09	6.15
Italy	6.54	18.82	1.10	12.49	8.00	11.45	-6.89	0.04	1.04
Japan	7.01	22.20	-13.32	14.70	21.53	13.13	-34.85	0.00	1.57
Malaysia	6.60	35.38	30.26	26.79	-19.30	16.23	49.56	0.00	10.56
Mexico	13.45	21.90	23.91	14.30	-9.77	10.82	33.68	0.00	3.48
Netherlands	11.39	18.38	7.52	12.97	5.48	10.05	2.04	0.52	2.92
Norway	5.34	28.61	-15.23	18.79	24.12	16.51	-39.35	0.00	2.28
Peru	22.81	15.16	25.71	7.52	-2.81	9.16	28.52	0.00	-1.64
Poland	26.70	64.19	34.63	33.94	-12.67	32.74	47.30	0.00	1.20
Russian Federati	40.84	30.26	46.43	22.33	13.37	14.90	33.05	0.01	7.43
Saudi Arabia	19.77	21.09	18.41	17.59	2.10	12.40	16.31	0.19	5.19
Singapore	6.54	23.57	7.26	17.62	4.34	13.25	2.92	0.54	4.37
South Africa	13.14	24.88	32.56	14.63	-17.48	13.15	50.04	0.00	1.48
South Korea	14.63	35.50	14.40	24.47	6.10	21.28	8.30	0.20	3.19
Spain	12.19	18.72	6.05	13.81	8.47	10.81	-2.42	0.51	3.00
Switzerland	10.94	16.16	20.84	10.96	-8.37	8.46	29.20	0.00	2.50
Taiwan	5.17	24.35	-20.52	19.54	29.03	16.16	-49.55	0.00	3.38
Thailand	5.73	27.79	-2.22	20.81	10.67	15.62	-12.90	0.02	5.19
Turkey	24.74	42.44	-7.66	33.51	32.96	16.53	-40.63	0.00	16.98
United Kingdom	8.34	17.47	3.29	12.30	7.03	8.36	-3.74	0.16	3.94
United States	14.14	18.06	12.64	15.17	5.96	7.87	6.68	0.07	7.30
Developed Countries	11.66	17.41	12.02	11.33	3.94	10.06	8.08	0.00	1.27
Developing Countries	11.32	14.39	9.17	8.80	5.43	8.26	3.75	0.07	0.54

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Table 2: Portfolio Sorts

The table shows the annualized mean of the total return, day return and night return (in percentage) of quintile portfolios sorted on various firm characteristics (from Low to High). The last two columns show the annualized mean return of spread portfolios short on portfolio 1 and long on portfolio 5 (High-Low) and *p*-values of tests whether their mean return differs from zero. We annualize returns by multiplying daily returns by 252. To compute the table entries, we first form equally- (Panel A) or value-weighted (Panel B) portfolios by country and then average the returns of the portfolios across countries. The portfolios are formed in such a way that the spread portfolio is expected to generate a positive return. The definitions of the anomaly variables are provided in Section 3.1.

Panel A: Equally-Weighted Portfolios

	Raw Returns						
	Low	2	3	4	High	High-Low	*p*-value
Size							
Total Return	11.76	10.62	10.40	11.21	15.72	3.96	(0.01)
Day Return	-2.56	-0.21	3.48	11.19	26.95	29.52	(0.00)
Night Return	15.04	12.43	9.49	3.90	-4.04	-19.08	(0.00)
Book/Market							
Total Return	6.73	9.20	11.74	14.28	17.82	11.09	(0.00)
Day Return	-4.01	2.43	8.36	12.62	19.14	23.15	(0.00)
Night Return	13.48	9.44	6.37	4.74	2.99	-10.50	(0.00)
Momentum							
Total Return	5.28	9.52	12.20	14.63	18.06	12.78	(0.00)
Day Return	-3.14	8.00	10.06	11.90	11.77	14.92	(0.00)
Night Return	13.59	4.83	4.71	5.05	8.78	-4.81	(0.00)
Long-term Reversals							
Total Return	8.85	12.38	12.73	14.55	15.16	6.31	(0.00)
Day Return	0.81	8.78	10.63	12.24	13.59	12.78	(0.00)
Night Return	10.47	5.98	4.64	5.36	6.37	-4.10	(0.00)
Accruals							
Total Return	9.76	12.11	14.08	13.60	12.98	3.22	(0.00)
Day Return	5.00	7.66	11.45	9.52	6.73	1.73	(0.01)
Night Return	8.15	7.15	5.29	6.63	9.93	1.77	(0.00)
Net Share Issues							
Total Return	8.51	11.22	12.93	13.45	13.14	4.63	(0.00)
Day Return	-0.35	7.29	10.16	9.54	9.56	9.91	(0.00)
Night Return	12.26	7.25	5.45	6.30	5.94	-6.32	(0.00)
Asset Growth							
Total Return	9.17	12.49	13.24	14.41	12.82	3.65	(0.00)
Day Return	0.05	7.99	9.72	13.24	10.08	10.03	(0.00)
Night Return	11.84	7.00	5.98	4.17	7.09	-4.74	(0.00)
Profitability							
Total Return	10.54	12.09	11.70	13.07	12.22	1.68	(0.09)
Day Return	5.71	10.03	7.73	8.44	6.46	0.75	(0.37)
Night Return	10.48	5.59	6.45	6.76	7.92	-2.56	(0.00)

(continued)

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Table 2: Portfolio Sorts (continued)

Panel B: Value-Weighted Portfolios

	Raw Returns						
	Low	2	3	4	High	High-Low	*p*-value
Size							
Total Return	11.40	10.42	9.96	9.58	9.90	-1.49	(0.37)
Day Return	-0.65	1.07	4.50	10.00	18.10	18.75	(0.00)
Night Return	12.44	10.82	7.80	3.09	-2.44	-14.87	(0.00)
Book/Market							
Total Return	9.25	10.34	12.18	13.77	14.89	5.64	(0.00)
Day Return	-2.25	-0.87	0.98	3.71	6.21	8.46	(0.00)
Night Return	12.13	11.82	11.95	11.18	10.51	-1.61	(0.05)
Momentum							
Total Return	3.38	8.81	10.21	13.50	16.08	12.70	(0.00)
Day Return	-14.01	-2.05	0.77	3.57	2.21	16.22	(0.00)
Night Return	18.91	11.81	10.12	10.61	14.66	-4.26	(0.00)
Long-term Reversals							
Total Return	8.69	12.36	12.58	12.70	12.68	3.99	(0.02)
Day Return	-4.94	1.43	3.96	3.86	1.09	6.03	(0.00)
Night Return	14.26	11.47	9.47	9.85	13.21	-1.05	(0.26)
Accruals							
Total Return	9.10	11.27	11.95	12.19	11.92	2.81	(0.01)
Day Return	-4.80	-0.95	4.55	2.31	-1.79	3.01	(0.00)
Night Return	14.87	12.89	8.38	10.36	14.57	-0.30	(0.61)
Net Share Issues							
Total Return	8.78	10.47	11.28	13.01	12.05	3.27	(0.00)
Day Return	-5.79	-1.88	1.54	6.17	2.66	8.45	(0.00)
Night Return	15.33	13.07	10.64	7.73	9.92	-5.41	(0.00)
Asset Growth							
Total Return	9.40	11.13	13.43	11.60	11.35	1.95	(0.12)
Day Return	-4.82	0.77	4.35	2.47	-1.49	3.33	(0.00)
Night Return	14.82	10.91	10.06	9.84	14.05	-0.77	(0.25)
Profitability							
Total Return	8.34	9.98	11.34	11.55	11.86	3.52	(0.01)
Day Return	-10.95	-0.12	0.45	0.44	2.04	12.99	(0.00)
Night Return	21.13	11.61	11.52	11.66	10.32	-10.80	(0.00)

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Table 3: Fama-MacBeth Regressions

The table shows results of Fama-MacBeth regressions of total returns, day returns and night returns compounded over the next month on firm characteristics. Panel A shows results for all countries (World), the United States, and all countries excluding the United States. Panel A shows results for all countries (World), the United States, and all countries outside the United States. Panel B shows results separately for the eight most important non-U.S. countries. Definitions of the firm characteristic variables are provided in Section 3.1. Risk premia estimates are per month and in percentage. We calculate *t*-statistics using the formula of Newey and West (1987) with a lag length of twelve months. "Average # Firms" is the time-series mean of the cross-country average of the number of firms per (country-specific) cross-sectional regression. "Average R-Squared" is the time-series mean of the cross-country average of the R-squared per (country-specific) cross-sectional regression. "Time-Series Observations" is the total number of months used in a test, and "Total Observation" the total number of observations.

Panel A: World, United States, and Non-US. samples

	World						United States						Non-United States					
	Total		Day		Night		Total		Day		Night		Total		Day		Night	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
Size	-0.044	[-1.57]	-0.807	[-8.16]	0.343	[9.61]	0.004	[0.09]	-0.680	[-8.71]	0.230	[4.46]	-0.047	[-1.60]	-0.813	[-7.80]	0.348	[9.14]
Book-to-Market	0.208	[2.31]	0.466	[3.51]	-0.355	[-6.49]	0.112	[1.15]	0.289	[2.67]	-0.274	[-4.37]	0.212	[2.27]	0.475	[3.45]	-0.359	[-6.35]
Momentum	0.004	[1.11]	0.019	[4.03]	-0.014	[-3.75]	0.002	[0.48]	-0.008	[-2.81]	0.005	[2.72]	0.004	[1.09]	0.020	[4.07]	-0.015	[-3.92]
LT Reversals	-0.002	[-1.97]	0.001	[0.79]	-0.002	[-2.68]	-0.002	[-2.89]	-0.004	[-5.86]	0.001	[2.92]	-0.002	[-1.87]	0.001	[1.03]	-0.003	[-2.80]
Net Accruals	-0.295	[-1.85]	0.323	[0.85]	-0.016	[-0.09]	0.026	[0.29]	-0.177	[-1.40]	0.226	[2.27]	-0.320	[-1.87]	0.357	[0.88]	-0.030	[-0.15]
Shares Issuances	-0.746	[-3.12]	-2.008	[-3.56]	1.092	[2.96]	-1.858	[-5.90]	-4.618	[-8.78]	3.346	[7.14]	-0.692	[-2.85]	-1.923	[-3.22]	1.011	[2.60]
Asset Growth	-0.333	[-1.92]	-1.440	[-3.37]	0.640	[3.28]	-0.145	[-1.09]	-0.780	[-5.01]	0.352	[2.31]	-0.335	[-1.81]	-1.485	[-3.27]	0.660	[3.21]
Profitability	0.655	[2.47]	1.083	[2.38]	-1.150	[-3.03]	0.245	[3.24]	0.692	[6.25]	-0.685	[-8.47]	0.669	[2.42]	1.107	[2.31]	-1.180	[-2.94]
Average # Firms	385		385		385		2,789		2,789		2,789		259		259		259	
Average R-Squared	19.51%		18.02%		17.89%		2.23%		2.30%		1.40%		20.44%		18.86%		18.80%	
Time-Series Observations	228		228		228		228		228		228		228		228		228	
Total Observations	2,076,547		2,076,547		2,076,547		635,961		635,961		635,961		1,440,586		1,440,586		1,440,586	

(continued)

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Table 3: Fama-MacBeth Regressions (continued)

Panel B: Eight Most Important Non-U.S. Countries

	Australia		Canada		France		Germany		Hong Kong		Italy		Japan		United Kingdom	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
Total Returns																
Size	-0.059	[-0.85]	-0.180	[-2.02]	0.054	[1.14]	0.013	[0.23]	-0.064	[-0.51]	-0.123	[-1.48]	0.006	[0.09]	0.019	[0.25]
Book-to-Market	0.621	[2.82]	-0.077	[-0.94]	0.195	[2.00]	0.234	[2.36]	0.004	[0.03]	0.206	[1.94]	0.367	[4.01]	0.418	[1.71]
Momentum	0.012	[3.53]	0.006	[1.40]	0.011	[2.53]	0.015	[3.04]	0.004	[0.76]	0.011	[2.01]	-0.006	[-1.35]	0.006	[1.19]
LT Reversals	-0.001	[-1.35]	-0.001	[-1.36]	-0.003	[-2.13]	-0.002	[-0.76]	-0.001	[-0.33]	-0.002	[-0.80]	-0.006	[-2.42]	-0.002	[-2.04]
Net Accruals	0.119	[0.42]	-0.595	[-1.15]	-0.031	[-0.14]	0.390	[1.68]	-0.190	[-0.49]	-0.097	[-0.74]	-0.177	[-1.74]	-0.315	[-1.49]
Shares Issuances	-0.708	[-2.09]	-1.264	[-2.64]	-0.793	[-2.42]	-0.858	[-2.12]	-0.071	[-0.14]	-1.100	[-1.53]	0.388	[0.33]	-0.756	[-2.09]
Asset Growth	-0.382	[-1.39]	-0.771	[-2.23]	-0.336	[-1.15]	-0.868	[-2.67]	-0.158	[-0.46]	-0.476	[-1.30]	0.185	[0.43]	-0.556	[-1.64]
Profitability	0.630	[1.88]	0.255	[1.00]	0.455	[1.25]	0.340	[1.48]	0.429	[0.65]	0.324	[0.59]	0.347	[1.22]	0.560	[2.31]
Average # Firms	233		387		262		298		192		112		1,699		268	
Average R-Squared	11.21%		7.68%		9.14%		6.52%		13.53%		14.35%		6.70%		9.77%	
Time-Series Observations	228		228		228		144		192		216		228		228	
Total Observations	53,053		88,132		59,647		42,927		36,798		24,121		387,410		61,206	61,206
Day Returns																
Size	0.309	[4.12]	-1.906	[-2.20]	-0.397	[-3.93]	-1.336	[-4.24]	-2.032	[-5.85]	-0.571	[-4.63]	0.080	[0.81]	-0.39169	-4.22
Book-to-Market	1.524	[5.94]	-1.084	[-1.29]	0.226	[1.75]	1.290	[2.62]	0.783	[3.83]	0.675	[4.34]	0.445	[4.15]	0.516	[1.86]
Momentum	0.019	[4.75]	0.010	[1.18]	0.016	[3.64]	0.018	[1.90]	-0.013	[-2.53]	0.027	[4.10]	0.015	[2.63]	0.004	[0.83]
LT Reversals	-0.001	[-1.51]	0.000	[0.25]	-0.008	[-4.54]	0.001	[0.28]	-0.002	[-0.75]	-0.002	[-0.75]	0.001	[0.29]	-0.002	[-1.30]
Net Accruals	-0.451	[-1.07]	3.130	[1.10]	-0.369	[-1.20]	0.105	[0.19]	-0.502	[-1.07]	-0.075	[-0.48]	-0.085	[-0.59]	-0.082	[-0.35]
Shares Issuances	-3.293	[-3.98]	-9.861	[-2.71]	-3.333	[-6.43]	-1.108	[-1.77]	-1.097	[-1.73]	-3.841	[-3.29]	-0.345	[-0.41]	-0.938	[-2.28]
Asset Growth	-1.724	[-4.28]	-8.174	[-2.17]	-2.343	[-7.59]	-1.239	[-3.55]	-1.074	[-1.46]	-2.793	[-4.25]	0.830	[2.45]	-0.523	[-1.43]
Profitability	2.424	[3.85]	6.012	[1.68]	0.671	[1.23]	-0.216	[-0.41]	-0.387	[-0.43]	1.630	[2.11]	0.740	[2.57]	0.373	[1.20]
Average # Firms	233		387		262		298		192		112		1,699		268	
Average R-Squared	10.52%		6.23%		7.20%		8.25%		13.62%		12.64%		3.80%		8.24%	
Time-Series Observations	228		228		228		144		192		216		228		228	
Total Observations	53,053		88,132		59,647		42,927		36,798		24,121		387,410		61,206	
Night Returns																
Size	-0.52	[-7.56]	-0.04	[-0.49]	0.34	[4.93]	0.89	[4.70]	1.30	[6.00]	0.31	[2.81]	-0.20	[-2.31]	0.25	[3.68]
Book-to-Market	-0.93	[-4.56]	-0.19	[-1.06]	-0.04	[-0.27]	-0.59	[-4.21]	-0.69	[-4.67]	-0.41	[-3.84]	-0.08	[-0.87]	0.06	[0.46]
Momentum	-0.01	[-2.63]	-0.01	[-2.04]	-0.01	[-1.90]	-0.01	[-2.24]	0.01	[2.91]	-0.02	[-2.07]	-0.02	[-4.51]	0.00	[-1.11]
LT Reversals	0.00	[0.43]	0.00	[-0.93]	0.00	[2.62]	-0.01	[-2.43]	0.00	[0.22]	0.00	[0.05]	-0.01	[-3.85]	0.00	[-0.89]
Net Accruals	0.64	[1.50]	-0.27	[-0.70]	0.29	[1.06]	0.38	[1.63]	0.17	[0.40]	0.00	[-0.03]	-0.16	[-0.99]	-0.29	[-1.22]
Shares Issuances	2.97	[3.56]	3.35	[5.12]	2.28	[4.99]	0.56	[1.30]	0.94	[2.16]	2.94	[3.22]	0.74	[0.76]	0.22	[0.65]
Asset Growth	1.44	[3.65]	1.99	[4.85]	1.83	[6.05]	0.80	[2.71]	0.73	[1.38]	2.29	[3.72]	-0.81	[-2.72]	0.06	[0.29]
Profitability	-2.32	[-4.19]	-2.01	[-2.84]	-0.55	[-1.59]	-0.39	[-1.58]	0.32	[0.43]	-1.69	[-2.00]	-0.54	[-1.33]	-0.32	[-1.56]
Average # Firms	233		387		262		298		192		112		1,699		268	
Average R-Squared	10.75%		6.52%		6.57%		8.58%		15.66%		12.47%		3.97%		6.56%	
Time-Series Observations	228		228		228		144		192		216		228		228	
Total Observations	53,053		88,132		59,647		42,927		36,798		24,121		387,410		61,206	

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